Optimal Stopping with Memory

Deep recurrent policies for American options under rough volatility and multi-market backtesting

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MAC F244 Stochastic Calculus

American options are optimal stopping problems

- Exercising at date j pays $g_j = e^{-rt_j}(K - S_j)^+$; the price is $V_0 = \sup_\tau \mathbb{E}^{\mathbb{Q}}[g_\tau]$ over stopping times τ ($\{\tau = j\} \in \mathcal{F}_j$: decide using the past only).
- **Snell envelope:** $U_H = g_H, \;\; U_j = \max\big(g_j,\; \mathbb{E}[U_{j+1} \mid \mathcal{F}_j]\big)$.
  - smallest supermartingale above g; $\;\tau^* = \min\{j : U_j = g_j\}, \;\; V_0 = U_0$
- Exercise when the payoff beats the *continuation value* $C_j = \mathbb{E}[U_{j+1} \mid \mathcal{F}_j]$.
- Markov prices: $C_j = c_j(S_j)$, so tree, Longstaff–Schwartz and Deep Optimal Stopping only need today's state. **With memory (rough volatility) they do not.**

Continuous time: a free-boundary problem

- $\min\big(-\partial_t V - \mathcal{L}V,\ V - g\big) = 0$
  - continue: Black–Scholes / Feynman–Kac PDE
  - exercise: $V = K - S$
  - boundary $L(t)$: smooth pasting $\partial_S V = -1$
- Perpetual put: $L^* = \frac{2rK}{2r+\sigma^2} = 30$, $V(36) = 5.79$
- Finite T: Crank–Nicolson in $\log S$ + penalty for $V \geq g$
- **Optional stopping:** for a martingale no rule beats holding, so memory can only help when increments are dependent.

Boundary, $S_0 = 36$, $\sigma = 0.2$, $T = 1$:

t	$L(t)$
0.00	32.94
0.25	33.34
0.50	33.99
0.75	35.06
0.95	37.05
T^-	40.00

Ground truth: tree and PDE agree to 4×10^{-4}

S_0	σ	T	European	Bermudan (50/yr)		American	
				tree	PDE	tree	PDE
36	0.2	1	3.844	4.478	4.478	4.487	4.487
36	0.2	2	3.763	4.840	4.840	4.848	4.848
36	0.4	1	6.711	7.101	7.101	7.109	7.109
36	0.4	2	7.700	8.507	8.507	8.514	8.514
38	0.2	1	2.852	3.250	3.250	3.257	3.257
38	0.2	2	2.991	3.745	3.745	3.751	3.751
38	0.4	1	5.834	6.148	6.148	6.155	6.154
38	0.4	2	6.979	7.668	7.668	7.675	7.675
40	0.2	1	2.066	2.314	2.314	2.320	2.319
40	0.2	2	2.356	2.885	2.884	2.890	2.890
40	0.4	1	5.060	5.312	5.312	5.318	5.318
40	0.4	2	6.326	6.917	6.917	6.923	6.923
42	0.2	1	1.465	1.617	1.617	1.621	1.621
42	0.2	2	1.841	2.212	2.212	2.217	2.217
42	0.4	1	4.379	4.583	4.582	4.588	4.588
42	0.4	2	5.736	6.244	6.244	6.250	6.250
44	0.2	1	1.017	1.110	1.110	1.113	1.113
44	0.2	2	1.429	1.690	1.690	1.693	1.693
44	0.4	1	3.783	3.948	3.948	3.953	3.953
44	0.4	2	5.202	5.641	5.641	5.647	5.647

Baselines (P2)

- **Longstaff–Schwartz:** regress realised cash flows on $[1, x, x^2]$ of today's features, in-the-money paths only, backwards in time.
- **Deep Optimal Stopping:** one small network per date, trained backwards, each warm-started from the next date.
- **Fitted Q-iteration:** one network (MLP or GRU) for every date, regressed on bootstrapped targets $\max(g_{j+1}, K_{j+1})$.
- All see the same features; all prices measured out of sample (lower bounds).

OSPG: learn the stopping rule directly

- Policy outputs $\varphi_j = \mathbb{P}(\text{stop at } j \mid \text{not yet}, \ \mathcal{F}_j)$ from the history (GRU) or today only (MLP).
- Stopping does not change the path, so the stopping-time law is exact:

$$\psi_j = \varphi_j \prod_{n<j}(1-\varphi_n), \qquad J(\theta) = \mathbb{E}\Big[\sum_j \psi_j g_j\Big].$$

- Policy gradient with a constant baseline: $\nabla_\theta J = \mathbb{E}\big[\sum_j (g_j - b)\psi_j \nabla_\theta \log \psi_j\big]$.
- No rollouts, **no backward recursion**: no bias carried from date to date.
- Implicit Snell comparison: φ_j rises when $\dfrac{\psi_j g_j}{\sum_{n>j}\psi_n g_n} > \dfrac{\varphi_j}{1-\varphi_j}$.

What it took to make it work

- **Start mostly waiting** (output bias -2): from $\varphi \approx \frac{1}{2}$ the put policy collapsed to "stop now" (4.0 instead of 4.478).
- **Small penalty on the logits**: the gradient is proportional to φ; without it the GRU stalled at "never stop" on the 50-asset max-call.
- **Date 0 decided separately**: every simulated path starts in the same state.
- Batch norm in the MLP (as in the paper), using training statistics at decision time.
- Tests check the loss gradient equals the policy gradient, and that no method looks ahead.

Replication: OSPG matches the exact prices

Method	Put	Max-call d=20	Max-call d=50	Geo-call d=7	Geo-call d=20
LSM	4.458 (0.004)	51.69 (0.20)	69.83 (0.32)	10.169 (0.012)	9.899 (0.011)
DOS	4.465 (0.007)	51.69 (0.16)	69.72 (0.29)	–	–
FQI-MLP	4.467 (0.011)	51.68 (0.12)	69.67 (0.21)	10.145 (0.068)	9.874 (0.056)
FQI-GRU	–	51.73 (0.14)	69.81 (0.37)	–	–
OSPG-MLP	4.473 (0.008)	51.80 (0.20)	69.93 (0.28)	10.255 (0.046)	9.989 (0.017)
OSPG-GRU	4.465 (0.015)	51.72 (0.21)	69.87 (0.25)	10.237 (0.065)	9.941 (0.030)
Reference	4.478	51.67	69.74	10.260	10.033
Paper, DNN-OSPG	–	51.02	69.06	10.252	10.010

- All methods within 1.6% of the exact price; **OSPG-MLP closest**: -0.12% (put), -0.05% and -0.44% (geometric call), vs -0.9 to -1.6% for LS and FQI.
- Paired on the same paths: OSPG-GRU beats LS and FQI on the geometric call ($z \geq 2.6$), but trails OSPG-MLP by 0.1–0.5%: memory costs a little when there is none.
- The paper's max-call values are ≈ 0.8 below Becker et al.'s lower bound; we match Becker.

P3 scope: simulate memory, learn when to stop

- **fBm experiment:** a controlled non-Markovian stopping problem with reward $R_j = W^h_{t_j}$ and objective $\sup_\tau \mathbb{E}[W^h_\tau]$.
- **Rough Bergomi:** simulate a stochastic-volatility price process whose variance depends on a Volterra Gaussian process.
- Compare memory-aware GRU policies with MLP policies and classical baselines.
- Our deliverables: simulation, covariance / martingale / option-price checks, numerical experiments and analysis.

Exact fBm samples on a discrete grid

- Covariance:
$$\mathbb{E}[W_t^h W_s^h] = \tfrac{1}{2}(t^{2h} + s^{2h} - |t-s|^{2h}).$$
- Build the grid covariance C, factorise $C = LL^\top$, draw $Z \sim N(0, I)$, and generate correlated samples $W = ZL^\top$; prepend $W_0 = 0$.
- Empirical check at $h = 0.3$: terminal variance 1.0022 vs theoretical 1.0000; maximum absolute covariance error 0.00792 (40,000 paths, 10 intervals).
- Stopping study: 20,000 paths, 100 intervals, two random splits; 11 Hurst values from 0.05 to 0.95; DOS, FQI-MLP/GRU and OSPG-MLP/GRU.

fBm stopping results: reduced-compute run

h	DOS	FQI-MLP	FQI-GRU	OSPG-MLP	OSPG-GRU
0.05	0.700	0.910	1.297	1.153	1.286
0.10	0.508	0.740	1.040	0.926	1.032
0.20	0.304	0.481	0.643	0.578	0.639
0.30	0.187	0.299	0.366	0.334	0.362
0.40	0.092	0.132	0.158	0.141	0.151
0.50	-0.001	-0.002	-0.006	-0.004	-0.008
0.60	0.070	0.081	0.099	0.096	0.094
0.70	0.165	0.177	0.202	0.191	0.193
0.80	0.245	0.258	0.260	0.266	0.262
0.90	0.302	0.306	0.282	0.314	0.306
0.95	0.353	0.346	0.328	0.357	0.349

- At $h = 0.5$, all methods are close to zero, consistent with the Brownian martingale benchmark.
- GRU is competitive, but does not outperform MLP uniformly in this reduced run.
- Means are over two split-level estimates; interpret rankings cautiously.

Rough Bergomi: model and hybrid simulation

$$dS_t = rS_t\,dt + \sqrt{v_t}S_t\,dZ_t, \qquad dZ_t = \rho\,dW_t + \sqrt{1-\rho^2}\,dW_t^{\perp},$$
$$v_t = \xi_0 \exp\left(\eta\widetilde{W}_t^h - \tfrac{1}{2}\eta^2 t^{2h}\right),$$
$$\widetilde{W}_t^h = \sqrt{2h}\int_0^t (t-s)^{h-1/2}\,dW_s.$$

- Hybrid scheme: conditionally sample the near-diagonal kernel integral and approximate the remaining Volterra terms with hybrid evaluation weights.
- Log-Euler stock update uses the left-point variance; put reward is $e^{-rt_j}(K - S_{t_j})^+$.
- Preliminary European call check: 0.07886 ± 0.00061 (MC standard error), against reference 0.0791; difference is about 0.39 standard errors.

E4: American put under rough Bergomi

h	LSM	DOS	OSPG-MLP	OSPG-GRU
0.07	7.935 (0.340)	7.985 (0.350)	8.136 (0.340)	7.932 (0.356)
0.10	7.698 (0.329)	7.776 (0.348)	7.934 (0.340)	7.690 (0.359)
0.30	7.362 (0.337)	7.435 (0.350)	7.507 (0.332)	7.367 (0.368)
0.50	7.540 (0.314)	7.619 (0.315)	7.605 (0.352)	7.479 (0.360)

- 3,000 paths, 25 exercise intervals and one random split per h.
- OSPG-MLP has the largest point estimate in these pilot runs, but the standard errors overlap substantially; no method is shown to be statistically superior.
- Main takeaway: the simulators and method interfaces run end-to-end; larger multi-split experiments are needed for stronger performance claims.

Real market data and reproducible windows (P4)

- Adjusted equity closes: 50 Indian and 100 usable US tickers; Binance spot: eight crypto assets.
- Frozen 2015–2024 equity history; crypto requested from 2017. Retrieval-date universes have selection bias.
- Audited cleaning: no future filling; missing windows excluded; large moves flagged and retained.
- 30-calendar-day deadline; five-observation entry stride; 72,048 windows before fold selection.
- Causal return, drawdown, volatility, lagged returns, remaining time; lagged local index/VIX context.

Execution, costs and fair comparisons (P4)

$$r_j = \frac{S_j}{S_0}\frac{1 - c_s}{1 + c_b} - 1 - fd_j$$

- Daily-close reference-price experiment; guaranteed same-close execution is not assumed.
- Fees, half-spread and slippage on both legs; configurable research scenarios and cost sensitivity.
- Hold to deadline; fixed $+5\%/-3\%$ profit/loss rule; 5% trailing stop.
- Expanding 36-month train / 6-month validation / 6-month test folds; purge overlapping outcomes.
- Linear and exponential-utility rewards; actual net returns retained for both.

Validation and the statistics handoff (P4)

- 71 tests passed, including 12 P4 accounting, causality, split and integration tests.
- Real-data smoke verified LSM, DOS, FQI-MLP/GRU and OSPG-MLP/GRU interfaces.
- Saved asset/window IDs, entry/deadline/decision dates, stopping indices and actual exit dates.
- Full rule runs: 12 walk-forward folds per equity market; seven crypto folds.
- P5 performs final training and date-block inference: overlapping trades are dependent and are not portfolio returns.

Backtest results (P5)

- To do: results by asset class, bootstrap intervals, regimes, memory vs Hurst.